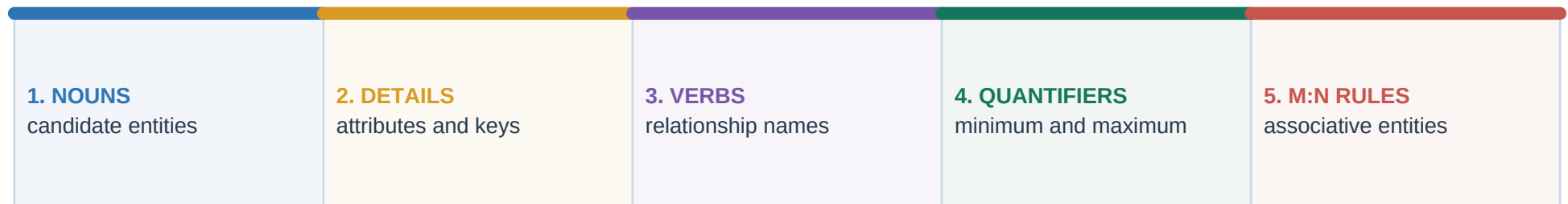


FROM BUSINESS RULES TO A COMPLETE ERD

A step-by-step visual surgery of July 2026 Question 5



Worked case: Student Merit Activities Database System. The completed model contains six tables and resolves two many-to-many relationships.

Full business-rules question

July 2026 - Question 5 - 20 marks

Every semester, a large number of students apply to stay at the university hostel. Currently, the hostel management filters the applications by manually calculating the total merit points of each student. As the number of applications continues to increase, this process has become inefficient and time-consuming. Therefore, the hostel management has decided to develop a Student Merit Activities Database System to manage student activities and automatically calculate merit points for hostel applications. As a database designer, you are tasked with designing the database system according to the following requirements:

ENTITY	ATTRIBUTE	RELATIONSHIP	CARDINALITY	ASSOCIATIVE CLUE
1	The system must record student details, including ID, name, phone number, and program code . Extract: STUDENT entity and four attributes.			
2	Each student may participate in multiple events , and each event should be participated by at least one student. There is a unique ID for each participation . Extract: M:N Student-Event rule, PARTICIPATION associative entity, ParticipationID and both endpoint minima.			
3	Event details such as ID, name, date, and merit points are recorded. Extract: EVENT entity and four attributes.			
4	An event may require multiple pieces of equipment , and each piece of equipment may be used across multiple events. The system must track the total quantity of equipment used per event . Extract: M:N Event-Equipment rule, EVENT_EQUIPMENT associative entity and QuantityUsed.			
5	Each equipment has its ID, name, and total quantity in store . Extract: EQUIPMENT entity and stock attributes.			
6	Each event needs to be supervised by one staff member . There are some staff members who do not supervise any events . Finally, each event is led by one student , designated as the Event Leader. Extract: Staff-Event and Student-Event Leader relationships, with mandatory Event endpoints and optional Staff/Student endpoints.			
(a)	Construct a complete Entity Relationship Diagram (ERD) with all of the required tables, attributes, and relationships using the Crow's Foot notation that reflects all of the given business rules.			(16 marks)
(b)	Identify TWO relevant information outputs that can be generated from the Student Merit Activities Database System.			(4 marks)

Step 1 - Perform noun surgery

A noun is only a candidate entity. Keep it when the system stores several distinguishable occurrences and each occurrence needs its own facts.

Business-rule clue	What it reveals	Decision
Record student details: ID, name, phone and program.	An identifiable object with descriptive facts.	STUDENT entity
Event details: ID, name, date and merit points.	Many events, each with its own details.	EVENT entity
Each equipment item has an ID, name and stock quantity.	Reusable across events; not an Event attribute.	EQUIPMENT entity
Each event is supervised by one staff member.	Staff must be identifiable even though attributes are omitted.	STAFF ; add StaffID and StaffName

Boundary: Hostel application explains the business problem, but the supplied rules do not require an APPLICATION table. Do not model facts that were never requested.

Step 2 - Attach attributes and keys

STUDENT

StudentID, Name, PhoneNumber,
ProgramCode

EVENT

EventID, EventName, EventDate,
MeritPoints, *StaffID**,
*LeaderStudentID**

EQUIPMENT

EquipmentID, EquipmentName,
TotalQuantity

STAFF

StaffID, StaffName

Step 3 - Resolve Student participation

Read the rule in both directions. The words **may**, **multiple** and **at least one** determine the endpoints.

Rule fragment	Minimum / maximum	ERD result
A student may participate in multiple events.	One student: zero to many events.	STUDENT 1 : 0..M PARTICIPATION
Each event has at least one student.	One event: one to many students.	EVENT 1 : 1..M PARTICIPATION
There is a unique ID for each participation.	The pairing needs its own record.	Create PARTICIPATION

PARTICIPATION	<u>ParticipationID</u> , StudentID*, EventID*
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Why it works: PARTICIPATION converts the M:N relationship into two 1:M relationships and stores the question's unique participation identifier.

Step 4 - Resolve Event equipment

Phrase	Meaning	Placement
An event may require multiple equipment items.	EVENT to EQUIPMENT is 0..M.	Resolve the M:N relationship.
Equipment may be used across multiple events.	EQUIPMENT to EVENT is 0..M.	Create EVENT_EQUIPMENT.
Track quantity used per event.	QuantityUsed describes one pairing.	Store it in EVENT_EQUIPMENT.

Step 5 - Add supervision and leadership

STAFF supervises EVENT	Each EVENT has exactly one STAFF supervisor. A STAFF member may supervise zero or many events. Place StaffID as a foreign key in EVENT.
STUDENT leads EVENT	Each EVENT has exactly one STUDENT leader. A STUDENT may lead zero or many events. Place LeaderStudentID as a foreign key in EVENT.

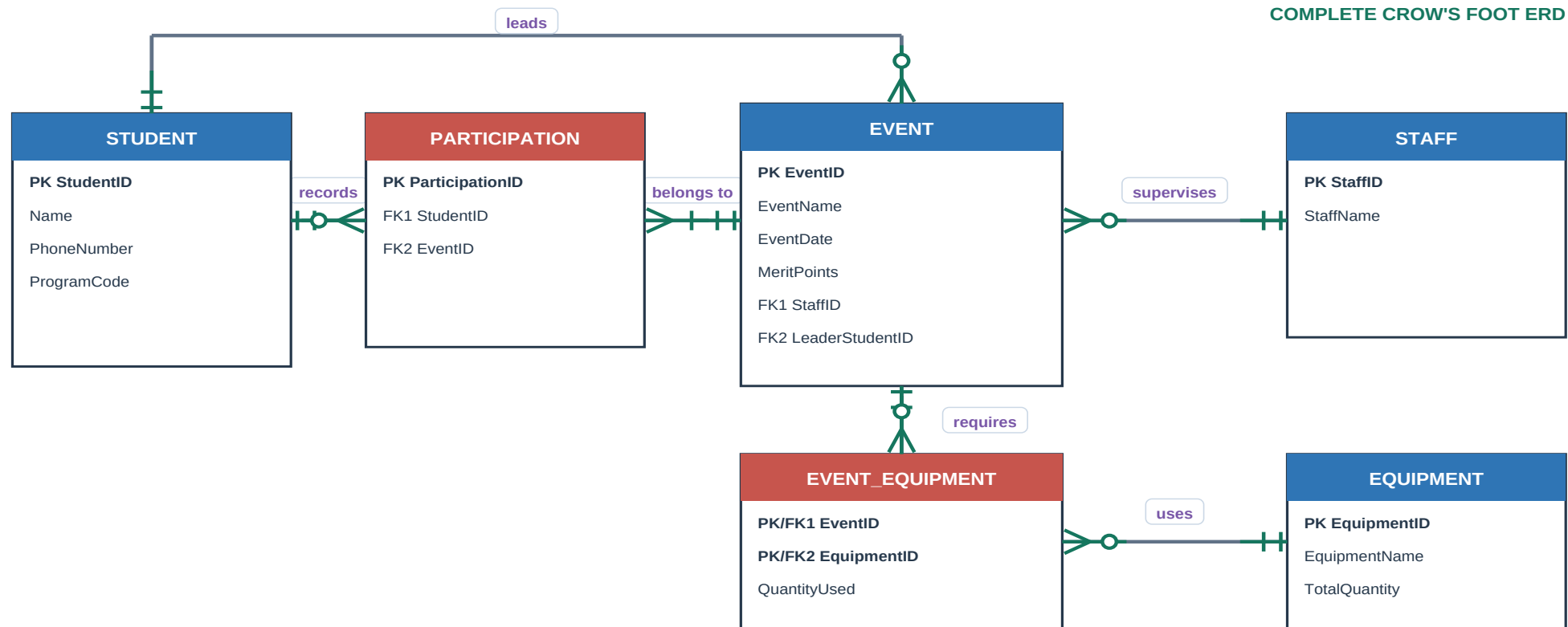
Do not invent this rule: the question does not say that the event leader must also be recorded as a participant. It may be a useful policy, but it is not part of the supplied scenario.

Step 6 - Translate quantifiers into endpoints

Wording	Minimum	Maximum	Crow's Foot endpoint
may / some do not	zero	read the rest of the phrase	O
each / one / exactly one	one	one	
at least one	one	many	<
multiple / many / several	use may or must	many	<

Step 7 - Assemble the complete ERD

The connected model below combines all entities, attributes, foreign keys, relationship names and minimum-maximum endpoints.



How to read the endpoints: the symbol beside a table describes how many rows of that table may match one row at the opposite end. For example, O< beside EVENT and || beside STAFF mean that one staff member may supervise zero or many events, while every event has exactly one staff supervisor. Red headers identify associative entities.

Final relational structure

STUDENT	<u>StudentID</u> , Name, PhoneNumber, ProgramCode
EVENT	<u>EventID</u> , EventName, EventDate, MeritPoints, <i>StaffID*</i> , <i>LeaderStudentID*</i>
PARTICIPATION	<u>ParticipationID</u> , <i>StudentID*</i> , <i>EventID*</i>
EQUIPMENT	<u>EquipmentID</u> , EquipmentName, TotalQuantity
EVENT_EQUIPMENT	<i>EventID*</i> , <i>EquipmentID*</i> , QuantityUsed
STAFF	<u>StaffID</u> , StaffName

Six tables Four main entities and two associative entities.	Every table has a key Including the composite key in EVENT_EQUIPMENT.
Both M:N rules are resolved Student-Event and Event-Equipment.	Every endpoint is justified Optionality follows words such as may and some do not.

How to write the examination answer

- Draw every entity and required attribute.
- Mark all primary and foreign keys clearly.
- Name each relationship with a meaningful verb.
- Show connectivity, minimum participation and maximum cardinality at both ends.
- Re-read every business rule and confirm the model can store each stated fact.